

Zhang-Style Forecast Evaluation for Dow 30, S&P 100, and S&P 500 GNNHAR Runs

Generated from local Volatility outputs

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Abstract

This report summarizes the post-run evidence generated from the saved Dow 30, S&P 100, and S&P 500 GNNHAR forecasts. The objective is not to retrain models, but to read the local prediction arrays and compute the statistics used in Zhang et al. (2024a)’s GNNHAR paper: HAR-normalized MSE and QLIKE loss ratios, model confidence sets (Hansen et al., 2011), calm-versus-turbulent regime tables, pairwise QLIKE Diebold-Mariano comparisons (Diebold and Mariano, 1995), FVU nonlinearity diagnostics, forecast error and forecast ratio boxplot sources, graph summaries, and smoothing diagnostics. The S&P 500 analysis is now tied to the actual AutoDL run source rather than the smaller Google Drive upload panel.

1 Notation

Let $v_{i,t}$ denote the realized variance or realized volatility target for ticker i on date t , and let $\hat{v}_{i,t}^{(m)}$ be the forecast from model m . The benchmark model is HAR_M , the HAR model estimated under MSE. The loss ratios are

$$R_m^{MSE} = \frac{\frac{1}{NT} \sum_{i,t} (v_{i,t} - \hat{v}_{i,t}^{(m)})^2}{\frac{1}{NT} \sum_{i,t} (v_{i,t} - \hat{v}_{i,t}^{(HAR_M)})^2}, \quad R_m^{QL} = \frac{\frac{1}{NT} \sum_{i,t} QL(v_{i,t}, \hat{v}_{i,t}^{(m)})}{\frac{1}{NT} \sum_{i,t} QL(v_{i,t}, \hat{v}_{i,t}^{(HAR_M)})}.$$

The QLIKE loss is

$$QL(y, \hat{y}) = \frac{y}{\hat{y}} - \log\left(\frac{y}{\hat{y}}\right) - 1.$$

A ratio below one means lower average forecast loss than HAR_M . $GHAR_kH$ denotes a linear graph-HAR model using k -hop neighbors; $GNNHAR_kL$ denotes a k -layer graph neural HAR model. The suffix $_IV$ indicates that lagged implied volatility is included. MCS denotes the Hansen–Lunde–Nason model confidence set at the 5% level. DM denotes the QLIKE Diebold–Mariano comparison; in the generated files, a positive statistic favors the candidate model listed in the comparison.

2 Data Source Alignment and the 397-versus-449 Issue

The earlier discrepancy came from comparing the S&P 500 AutoDL forecasts against the wrong raw panel. The formal AutoDL run used the local equivalent of `data/scale_experiment/sp500`, whose RV, IV, and return panels each contain 503 tickers before coverage filtering. The model run selected 449 tickers. The smaller 397-column RV/IV panel belongs to the Google Drive upload copy and is not the correct source for the formal S&P 500 run. After rebinding the analysis script to the AutoDL metadata source, all 449 model tickers match the S&P 500 RV, IV, and return panels.

Table 1: Run-level sample summary

Universe	Test dates	Tickers	Models	Test start	Test end	Raw source rule	RV missing
dow30	234	30	8	2025-07-07	2026-06-09	aligned_wide_multihop_rerun	0
sp100	234	91	32	2025-07-07	2026-06-09	run_config_data_paths_data_dir	0
sp500	223	449	36	2025-07-14	2026-06-01	autodl_run_metadata_args_data_dir	0

The table uses the current aligned sample where available. Dow 30 now refers to the 2025-07-07–2026-06-09 aligned wide-window multi-hop rerun; the older 232-date Dow 30 full-model run is archived evidence and should not be described as the current out-of-sample date range. The S&P 500 source rule is AutoDL run metadata.

Table 2: Ticker alignment audit

Universe	Panel	Source rule	Panel tickers	Model tickers	Matched	Missing	Extra
dow30	local_raw_rv_panel	run_config_data_paths_data_dir	30	30	30	0	0
dow30	local_raw_return_panel	run_config_data_paths_data_dir	30	30	30	0	0
dow30	local_raw_iv_panel	run_config_data_paths_data_dir	30	30	30	0	0
sp100	local_raw_rv_panel	run_config_data_paths_data_dir	91	91	91	0	0
sp100	local_raw_return_panel	run_config_data_paths_data_dir	101	91	91	0	10
sp100	local_raw_iv_panel	run_config_data_paths_data_dir	91	91	91	0	0
sp500	local_raw_rv_panel	autodl_run_metadata_args_data_dir	503	449	449	0	54
sp500	local_raw_return_panel	autodl_run_metadata_args_data_dir	503	449	449	0	54
sp500	local_raw_iv_panel	autodl_run_metadata_args_data_dir	503	449	449	0	54

Extra tickers are available in the raw panel but excluded by the modeling coverage filter; they are not missing model inputs.

3 Implemented Zhang-Style Statistics

Table 3: Coverage of Zhang-style statistics

Component	Paper reference	Status
forecast_loss_ratios	Table 1 and Table 5	implemented_from_saved_forecasts
weekly_monthly_horizons	Table 1, Table 2, Table 3, Table 5	scope_gap_current_saved_forecasts_are_one_day_only
model_confidence_set	Table 1, Table 2, Table 4, Table 5	implemented_from_per_ticker_losses
market_regime_losses	Table 2	implemented_from_saved_forecasts
alternative_validation_size	Table 4	scope_gap_requires_new_rolling_runs
forecast_error_ratio_boxplots	Figure 5	implemented_as_source_tables_and_pngs
fvu_nonlinearity	Table 3 and equation (11)	implemented_with_regime_split
multi_hop_dm	Figure 6 and Appendix E.1	implemented_when_models_exist
mad_smoothing	Figure 7 and equation (12)	implemented_when_graph_and_hidden_or_forecast_representations_exist
data_summary_and_correlation	Appendix A and Figure 1	implemented_for_available_local_panels
graph_structure	Appendix A.2	implemented_when_graph_matrices_exist

The current saved forecasts are one-day horizon forecasts. Zhang’s one-week and one-month tables require separate horizon-specific target construction and reruns. Exact hidden-state MAD requires saved hidden representations; current diagnostics include forecast-level smoothing proxies.

4 Main Loss-Ratio Evidence

Tables below report the complete model set available for each universe in the current one-day-horizon saved forecasts. These tables correspond to Zhang’s Table 1 and Table 5 style, with the added IV variants and the S&P 500 extension.

Table 4: Complete loss ratios for DOW30

Model	Family	Depth/hop	IV	Est.	MSE ratio	QL ratio	MCS MSE	MCS QL
GNNHAR2L.Q_IV	GNNHAR	L2	yes	QLIKE	0.892	0.889	yes	yes
GHAR.Q_IV	GHAR	H1	yes	QLIKE	0.908	0.909	no	no
GNNHAR3L.Q_IV	GNNHAR	L3	yes	QLIKE	0.916	0.912	no	no
GNNHAR1L.Q_IV	GNNHAR	L1	yes	QLIKE	0.924	0.914	no	no
GNNHAR2L.M_IV	GNNHAR	L2	yes	MSE	0.948	0.950	no	no
HAR.Q_IV	HAR		yes	QLIKE	0.998	0.957	no	no
GNNHAR2L.Q	GNNHAR	L2	no	QLIKE	0.967	0.957	no	no
GNNHAR3L.Q	GNNHAR	L3	no	QLIKE	0.997	0.971	no	no
GNNHAR1L.Q	GNNHAR	L1	no	QLIKE	0.986	0.980	no	no
HAR.M	HAR		no	MSE	1.000	1.000	no	no
GNNHAR1L.M	GNNHAR	L1	no	MSE	0.945	1.007	no	no
HAR.Q	HAR		no	QLIKE	1.090	1.013	no	no
GHAR.Q	GHAR	H1	no	QLIKE	0.959	1.021	no	no
HAR.M_IV	HAR		yes	MSE	0.920	1.030	no	no
GNNHAR3L.M	GNNHAR	L3	no	MSE	0.982	1.058	no	no
GNNHAR1L.M_IV	GNNHAR	L1	yes	MSE	0.935	1.076	no	no
GHAR.M_IV	GHAR	H1	yes	MSE	0.868	1.137	yes	no
GHAR.M	GHAR	H1	no	MSE	0.927	1.140	no	no
GNNHAR2L.M	GNNHAR	L2	no	MSE	0.983	1.222	no	no
GNNHAR3L.M_IV	GNNHAR	L3	yes	MSE	0.995	1.233	no	no

Ratios are relative to HAR_M . MCS columns indicate inclusion in the model confidence set based on per-ticker losses.

Table 5: Complete loss ratios for SP100

Model	Family	Depth/hop	IV	Est.	MSE ratio	QL ratio	MCS MSE	MCS QL
HAR.Q_IV	HAR		yes	QLIKE	0.981	0.933	yes	yes
GHAR.Q_IV	GHAR	H1	yes	QLIKE	0.981	0.933	yes	yes
GNNHAR2L.Q_IV	GNNHAR	L2	yes	QLIKE	0.982	0.933	no	yes
GNNHAR5L.Q_IV	GNNHAR	L5	yes	QLIKE	0.981	0.934	yes	yes
GNNHAR1L.Q_IV	GNNHAR	L1	yes	QLIKE	0.981	0.935	no	yes
GNNHAR4L.Q_IV	GNNHAR	L4	yes	QLIKE	0.986	0.936	no	yes
GNNHAR3L.Q_IV	GNNHAR	L3	yes	QLIKE	0.981	0.940	no	no
GHAR.Q	GHAR	H1	no	QLIKE	1.003	0.976	no	no
GNNHAR5L.Q	GNNHAR	L5	no	QLIKE	1.003	0.977	no	no
GHAR2H.M_IV	GHAR	H2	yes	MSE	0.965	0.977	yes	no
GNNHAR3L.Q	GNNHAR	L3	no	QLIKE	1.003	0.977	no	no
GHAR.M_IV	GHAR	H1	yes	MSE	0.965	0.978	yes	no
GHAR3H.M_IV	GHAR	H3	yes	MSE	0.965	0.978	yes	no
GNNHAR4L.Q	GNNHAR	L4	no	QLIKE	1.003	0.978	no	no
HAR.M_IV	HAR		yes	MSE	0.965	0.979	yes	no
GNNHAR2L.Q	GNNHAR	L2	no	QLIKE	1.004	0.979	no	no
HAR.Q	HAR		no	QLIKE	1.006	0.980	no	no
GNNHAR1L.Q	GNNHAR	L1	no	QLIKE	1.004	0.980	no	no
GNNHAR1L.M_IV	GNNHAR	L1	yes	MSE	0.972	0.987	yes	no
GNNHAR3L.M_IV	GNNHAR	L3	yes	MSE	0.968	0.990	yes	no
GHAR.M	GHAR	H1	no	MSE	0.997	0.993	no	no
GNNHAR5L.M_IV	GNNHAR	L5	yes	MSE	0.980	0.994	no	no
GNNHAR4L.M_IV	GNNHAR	L4	yes	MSE	0.979	0.996	no	no
GNNHAR1L.M	GNNHAR	L1	no	MSE	1.000	0.997	no	no
GHAR2H.M	GHAR	H2	no	MSE	0.998	0.998	no	no
GNNHAR2L.M	GNNHAR	L2	no	MSE	1.000	0.999	no	no
GNNHAR3L.M	GNNHAR	L3	no	MSE	1.001	1.000	no	no
HAR.M	HAR		no	MSE	1.000	1.000	no	no
GHAR3H.M	GHAR	H3	no	MSE	0.998	1.001	no	no
GNNHAR4L.M	GNNHAR	L4	no	MSE	1.004	1.003	no	no
GNNHAR5L.M	GNNHAR	L5	no	MSE	1.003	1.010	no	no
GNNHAR2L.M_IV	GNNHAR	L2	yes	MSE	0.989	1.012	no	no

Ratios are relative to HAR_M . MCS columns indicate inclusion in the model confidence set based on per-ticker losses.

Table 6: Complete loss ratios for SP500

Model	Family	Depth/hop	IV	Est.	MSE ratio	QL ratio	MCS MSE	MCS QL
GHAR_M_IV	GHAR	H1	yes	MSE	0.966	0.973	yes	yes
HAR_M_IV	HAR		yes	MSE	0.966	0.974	yes	no
GHAR2H_M_IV	GHAR	H2	yes	MSE	0.986	0.991	no	no
GHAR_M	GHAR	H1	no	MSE	1.000	0.998	no	no
GHAR2H_M	GHAR	H2	no	MSE	1.001	0.998	no	no
GHAR3H_M	GHAR	H3	no	MSE	1.001	0.998	no	no
HAR_M	HAR		no	MSE	1.000	1.000	no	no
GHAR3H_M_IV	GHAR	H3	yes	MSE	1.002	1.020	no	no
GHAR3H_Q_IV	GHAR	H3	yes	QLIKE	88.253	2.621	no	no
GHAR2H_Q_IV	GHAR	H2	yes	QLIKE	88.340	2.624	no	no
HAR_Q_IV	HAR		yes	QLIKE	124.370	2.677	no	no
GHAR_Q_IV	GHAR	H1	yes	QLIKE	83.373	2.698	no	no
GHAR_Q	GHAR	H1	no	QLIKE	161.079	2.851	no	no
GHAR3H_Q	GHAR	H3	no	QLIKE	142.460	2.895	no	no
GHAR2H_Q	GHAR	H2	no	QLIKE	142.009	2.899	no	no
HAR_Q	HAR		no	QLIKE	180.880	2.910	no	no
GNNHAR2L_Q_IV	GNNHAR	L2	yes	QLIKE	12.851	7.802	no	no
GNNHAR4L_Q_IV	GNNHAR	L4	yes	QLIKE	13.632	7.808	no	no
GNNHAR1L_Q_IV	GNNHAR	L1	yes	QLIKE	13.818	7.823	no	no
GNNHAR3L_Q_IV	GNNHAR	L3	yes	QLIKE	12.930	7.824	no	no
GNNHAR5L_Q_IV	GNNHAR	L5	yes	QLIKE	13.241	7.826	no	no
GNNHAR3L_Q	GNNHAR	L3	no	QLIKE	17.289	7.854	no	no
GNNHAR5L_Q	GNNHAR	L5	no	QLIKE	17.652	7.856	no	no
GNNHAR1L_Q	GNNHAR	L1	no	QLIKE	17.051	7.857	no	no
GNNHAR2L_Q	GNNHAR	L2	no	QLIKE	16.106	8.152	no	no
GNNHAR4L_Q	GNNHAR	L4	no	QLIKE	19.880	9.965	no	no
GNNHAR1L_M_IV	GNNHAR	L1	yes	MSE	6.261	10.912	no	no
GNNHAR5L_M	GNNHAR	L5	no	MSE	6.272	10.919	no	no
GNNHAR1L_M	GNNHAR	L1	no	MSE	6.282	10.922	no	no
GNNHAR3L_M_IV	GNNHAR	L3	yes	MSE	6.321	10.935	no	no
GNNHAR5L_M_IV	GNNHAR	L5	yes	MSE	6.361	10.951	no	no
GNNHAR2L_M_IV	GNNHAR	L2	yes	MSE	6.380	10.956	no	no
GNNHAR3L_M	GNNHAR	L3	no	MSE	6.466	10.987	no	no
GNNHAR4L_M_IV	GNNHAR	L4	yes	MSE	6.578	11.048	no	no
GNNHAR4L_M	GNNHAR	L4	no	MSE	6.743	11.098	no	no
GNNHAR2L_M	GNNHAR	L2	no	MSE	7.040	11.258	no	no

Ratios are relative to HAR_M . MCS columns indicate inclusion in the model confidence set based on per-ticker losses.

Table 7: Best models by universe and evaluation loss

Universe	Selection metric	Best model	Ratio	Gain	Family	IV
dow30	mse_ratio_vs_HAR_M	GHAR_M_IV	0.868	13.2%	GHAR	yes
dow30	qlike_ratio_vs_HAR_M	GNNHAR2L_Q_IV	0.889	11.1%	GNNHAR	yes
sp100	mse_ratio_vs_HAR_M	HAR_M_IV	0.965	3.5%	HAR	yes
sp100	qlike_ratio_vs_HAR_M	HAR_Q_IV	0.933	6.7%	HAR	yes
sp500	mse_ratio_vs_HAR_M	HAR_M_IV	0.966	3.4%	HAR	yes
sp500	qlike_ratio_vs_HAR_M	GHAR_M_IV	0.973	2.7%	GHAR	yes

The best QLIKE model differs by universe: Dow 30 favors a nonlinear IV model, S&P 100 favors HAR with IV, and S&P 500 favors GHAR with IV.

5 MCS and DM Interpretation

The MCS results are stricter than simple loss ranking. Dow 30 has a clear QLIKE winner $GNNHAR2L_Q_IV$. S&P 100 has a broad IV-trained QLIKE confidence set, indicating that several IV variants are statistically hard to separate. S&P 500 is much more linear in the current run: $GHAR_M_IV$ is the QLIKE MCS winner, while most QLIKE-trained neural specifications have very poor loss ratios.

Table 8: QLIKE DM comparison summary for DOW30

Type	Base	Candidate	Cand./base	Gain	pj0.05 share	Positive share
graph_linear	HAR_M	GHAR_M	1.140	-14.0%	0.0%	16.7%
graph_linear	HAR_M_IV	GHAR_M_IV	1.104	-10.4%	3.3%	30.0%
graph_linear	HAR_Q	GHAR_Q	1.008	-0.8%	6.7%	63.3%
graph_linear	HAR_Q_IV	GHAR_Q_IV	0.950	5.0%	20.0%	83.3%
nonlinear_one_hop	GHAR_M	GNNHAR1L_M	0.883	11.7%	10.0%	76.7%
nonlinear_one_hop	GHAR_M_IV	GNNHAR1L_M_IV	0.946	5.4%	0.0%	66.7%
nonlinear_one_hop	GHAR_Q	GNNHAR1L_Q	0.960	4.0%	0.0%	56.7%
nonlinear_one_hop	GHAR_Q_IV	GNNHAR1L_Q_IV	1.005	-0.5%	10.0%	40.0%
gnn_depth	GNNHAR1L_M	GNNHAR2L_M	1.214	-21.4%	56.7%	10.0%
gnn_depth	GNNHAR1L_M_IV	GNNHAR2L_M_IV	0.883	11.7%	43.3%	90.0%
gnn_depth	GNNHAR1L_Q	GNNHAR2L_Q	0.976	2.4%	13.3%	73.3%
gnn_depth	GNNHAR1L_Q_IV	GNNHAR2L_Q_IV	0.973	2.7%	23.3%	80.0%
gnn_depth	GNNHAR2L_M	GNNHAR3L_M	0.866	13.4%	13.3%	83.3%
gnn_depth	GNNHAR2L_M_IV	GNNHAR3L_M_IV	1.297	-29.7%	66.7%	3.3%
gnn_depth	GNNHAR2L_Q	GNNHAR3L_Q	1.015	-1.5%	10.0%	26.7%
gnn_depth	GNNHAR2L_Q_IV	GNNHAR3L_Q_IV	1.026	-2.6%	3.3%	30.0%

A candidate/base ratio below one means the candidate has lower average QLIKE than the base. Positive DM share is the share of tickers where the DM statistic favors the candidate.

Table 9: QLIKE DM comparison summary for SP100

Type	Base	Candidate	Cand./base	Gain	pj0.05 share	Positive share
graph_linear	HAR_M	GHAR_M	0.993	0.7%	22.0%	47.3%
graph_linear	HAR_M_IV	GHAR_M_IV	0.999	0.1%	12.1%	57.1%
graph_linear	HAR_Q	GHAR_Q	0.996	0.4%	6.6%	56.0%
graph_linear	HAR_Q_IV	GHAR_Q_IV	1.001	-0.1%	4.4%	50.5%
nonlinear_one_hop	GHAR_M	GNNHAR1L_M	1.004	-0.4%	15.4%	40.7%
nonlinear_one_hop	GHAR_M_IV	GNNHAR1L_M_IV	1.009	-0.9%	18.7%	44.0%
nonlinear_one_hop	GHAR_Q	GNNHAR1L_Q	1.004	-0.4%	15.4%	35.2%
nonlinear_one_hop	GHAR_Q_IV	GNNHAR1L_Q_IV	1.001	-0.1%	7.7%	40.7%
linear_multihop	GHAR2H_M	GHAR3H_M	1.003	-0.3%	33.0%	49.5%
linear_multihop	GHAR2H_M_IV	GHAR3H_M_IV	1.001	-0.1%	18.7%	39.6%
linear_multihop	GHAR_M	GHAR2H_M	1.006	-0.6%	20.9%	36.3%
linear_multihop	GHAR_M_IV	GHAR2H_M_IV	0.999	0.1%	16.5%	51.6%
gnn_depth	GNNHAR1L_M	GNNHAR2L_M	1.003	-0.3%	17.6%	45.1%
gnn_depth	GNNHAR1L_M_IV	GNNHAR2L_M_IV	1.026	-2.6%	18.7%	42.9%
gnn_depth	GNNHAR1L_Q	GNNHAR2L_Q	0.999	0.1%	3.3%	54.9%
gnn_depth	GNNHAR1L_Q_IV	GNNHAR2L_Q_IV	0.999	0.1%	14.3%	62.6%
gnn_depth	GNNHAR2L_M	GNNHAR3L_M	1.000	-0.0%	11.0%	58.2%
gnn_depth	GNNHAR2L_M_IV	GNNHAR3L_M_IV	0.978	2.2%	20.9%	54.9%
gnn_depth	GNNHAR2L_Q	GNNHAR3L_Q	0.998	0.2%	11.0%	54.9%
gnn_depth	GNNHAR2L_Q_IV	GNNHAR3L_Q_IV	1.007	-0.7%	15.4%	37.4%
gnn_depth	GNNHAR3L_M	GNNHAR4L_M	1.004	-0.4%	7.7%	31.9%
gnn_depth	GNNHAR3L_M_IV	GNNHAR4L_M_IV	1.006	-0.6%	15.4%	53.8%
gnn_depth	GNNHAR3L_Q	GNNHAR4L_Q	1.001	-0.1%	9.9%	46.2%
gnn_depth	GNNHAR3L_Q_IV	GNNHAR4L_Q_IV	0.996	0.4%	12.1%	61.5%
gnn_depth	GNNHAR4L_M	GNNHAR5L_M	1.007	-0.7%	4.4%	47.3%
gnn_depth	GNNHAR4L_M_IV	GNNHAR5L_M_IV	0.998	0.2%	11.0%	60.4%
gnn_depth	GNNHAR4L_Q	GNNHAR5L_Q	0.998	0.2%	13.2%	60.4%
gnn_depth	GNNHAR4L_Q_IV	GNNHAR5L_Q_IV	0.998	0.2%	8.8%	57.1%

A candidate/base ratio below one means the candidate has lower average QLIKE than the base. Positive DM share is the share of tickers where the DM statistic favors the candidate.

Table 10: QLIKE DM comparison summary for SP500

Type	Base	Candidate	Cand./base	Gain	pi0.05 share	Positive share
graph_linear	HAR_M	GHAR_M	0.998	0.2%	17.6%	51.2%
graph_linear	HAR_M_IV	GHAR_M_IV	0.999	0.1%	26.9%	56.1%
graph_linear	HAR_Q	GHAR_Q	0.980	2.0%	70.6%	61.7%
graph_linear	HAR_Q_IV	GHAR_Q_IV	1.008	-0.8%	65.0%	40.3%
nonlinear_one_hop	GHAR_M	GNNHAR1L_M	10.943	-994.3%	88.2%	4.2%
nonlinear_one_hop	GHAR_M_IV	GNNHAR1L_M_IV	11.211	-1021.1%	92.2%	1.8%
nonlinear_one_hop	GHAR_Q	GNNHAR1L_Q	2.756	-175.6%	86.4%	25.2%
nonlinear_one_hop	GHAR_Q_IV	GNNHAR1L_Q_IV	2.899	-189.9%	87.8%	20.9%
linear_multihop	GHAR2H_M	GHAR3H_M	1.000	-0.0%	11.8%	49.2%
linear_multihop	GHAR2H_M_IV	GHAR3H_M_IV	1.030	-3.0%	58.1%	8.0%
linear_multihop	GHAR2H_Q	GHAR3H_Q	0.999	0.1%	45.0%	60.4%
linear_multihop	GHAR2H_Q_IV	GHAR3H_Q_IV	0.999	0.1%	73.3%	52.1%
linear_multihop	GHAR_M	GHAR2H_M	1.000	-0.0%	13.4%	48.6%
linear_multihop	GHAR_M_IV	GHAR2H_M_IV	1.018	-1.8%	25.2%	28.3%
linear_multihop	GHAR_Q	GHAR2H_Q	1.017	-1.7%	47.0%	33.2%
linear_multihop	GHAR_Q_IV	GHAR2H_Q_IV	0.972	2.8%	61.2%	75.7%
gnn_depth	GNNHAR1L_M	GNNHAR2L_M	1.031	-3.1%	39.4%	36.7%
gnn_depth	GNNHAR1L_M_IV	GNNHAR2L_M_IV	1.004	-0.4%	29.2%	38.8%
gnn_depth	GNNHAR1L_Q	GNNHAR2L_Q	1.038	-3.8%	63.9%	23.6%
gnn_depth	GNNHAR1L_Q_IV	GNNHAR2L_Q_IV	0.997	0.3%	58.8%	49.0%
gnn_depth	GNNHAR2L_M	GNNHAR3L_M	0.976	2.4%	39.4%	46.5%
gnn_depth	GNNHAR2L_M_IV	GNNHAR3L_M_IV	0.998	0.2%	23.2%	44.3%
gnn_depth	GNNHAR2L_Q	GNNHAR3L_Q	0.963	3.7%	63.7%	70.6%
gnn_depth	GNNHAR2L_Q_IV	GNNHAR3L_Q_IV	1.003	-0.3%	52.3%	39.0%
gnn_depth	GNNHAR3L_M	GNNHAR4L_M	1.010	-1.0%	36.5%	45.0%
gnn_depth	GNNHAR3L_M_IV	GNNHAR4L_M_IV	1.010	-1.0%	34.7%	35.6%
gnn_depth	GNNHAR3L_Q	GNNHAR4L_Q	1.269	-26.9%	84.6%	7.6%
gnn_depth	GNNHAR3L_Q_IV	GNNHAR4L_Q_IV	0.998	0.2%	42.5%	56.8%
gnn_depth	GNNHAR4L_M	GNNHAR5L_M	0.984	1.6%	40.1%	51.0%
gnn_depth	GNNHAR4L_M_IV	GNNHAR5L_M_IV	0.991	0.9%	33.4%	50.3%
gnn_depth	GNNHAR4L_Q	GNNHAR5L_Q	0.788	21.2%	84.2%	88.4%
gnn_depth	GNNHAR4L_Q_IV	GNNHAR5L_Q_IV	1.002	-0.2%	48.6%	34.3%

A candidate/base ratio below one means the candidate has lower average QLIKE than the base. Positive DM share is the share of tickers where the DM statistic favors the candidate.

6 Regime, FVU, and Graph Diagnostics

Table 11: Best regime-specific loss ratios

Universe	Regime	Metric	Best model	Ratio	Gain	Dates
dow30	calm_bottom_90pct	MSE	GHAR_M_IV	0.973	2.7%	208
dow30	calm_bottom_90pct	QLIKE	GNNHAR2L_Q_IV	0.947	5.3%	208
dow30	turbulent_top_10pct	MSE	GHAR_M_IV	0.766	23.4%	24
dow30	turbulent_top_10pct	QLIKE	GHAR_M_IV	0.664	33.6%	24
sp100	calm_bottom_90pct	MSE	GHAR3H_M_IV	0.969	3.1%	210
sp100	calm_bottom_90pct	QLIKE	GNNHAR2L_Q_IV	0.933	6.7%	210
sp100	turbulent_top_10pct	MSE	HAR_M_IV	0.929	7.1%	24
sp100	turbulent_top_10pct	QLIKE	HAR_Q_IV	0.921	7.9%	24
sp500	calm_bottom_90pct	MSE	HAR_M_IV	0.968	3.2%	200
sp500	calm_bottom_90pct	QLIKE	GHAR_M_IV	0.976	2.4%	200
sp500	turbulent_top_10pct	MSE	GHAR_M_IV	0.944	5.6%	23
sp500	turbulent_top_10pct	QLIKE	GHAR_M_IV	0.943	5.7%	23

The current regime split uses the cross-sectional mean RV proxy because SPY is not present in these saved truth arrays or raw panels. Zhang uses SPY RV, so this is method-aligned but not data-identical.

Table 12: Lowest FVU relative to HAR_M by universe and regime

Universe	Regime	Model	Mean FVU	Median FVU	IV	Est.
dow30	calm_bottom_90pct	HAR_M	0.0000	0.0000	no	MSE
dow30	turbulent_top_10pct	HAR_M	0.0000	0.0000	no	MSE
sp100	calm_bottom_90pct	HAR_M	0.0000	0.0000	no	MSE
sp100	turbulent_top_10pct	HAR_M	0.0000	0.0000	no	MSE
sp500	calm_bottom_90pct	HAR_M	0.0000	0.0000	no	MSE
sp500	turbulent_top_10pct	HAR_M	0.0000	0.0000	no	MSE

FVU is an effect-size diagnostic for how far a model’s forecast function moves away from HAR_M ; it is not a significance test.

Table 13: Available graph rollout

Universe	Loss group	Graphs	Nodes	Edges	Density	Mean degree	Diameter
sp500	MSE	11	449.0	2238.8	0.022	9.97	10.73
sp500	QLIKE	11	449.0	2238.8	0.022	9.97	10.73

Only the S&P 500 AutoDL package currently contains graph matrices in the paper-ready tree; Dow 30 and S&P 100 forecasts remain analyzable, but graph structural summaries require saved graph matrices.

7 Comparison with Zhang et al.

Zhang et al. (2024a) find that GHAR improves on HAR, GNNHAR1L or GNNHAR2L often improves further, QLIKE training is useful especially under turbulent regimes, and very deep GNNHAR can deteriorate because of over-smoothing. Our Dow 30 result is directionally close once IV is included: GNNHAR2L_Q_IV is the best QLIKE model and improves roughly 11.1% over HAR_M . The S&P 100 run is weaker for graph nonlinearity: IV helps, but the best QLIKE model is HAR_Q_{IV} , and several IV graph/neural variants are statistically close by MCS. The S&P 500 AutoDL run is different again. It is internally consistent over 449 selected tickers, but the best QLIKE model is GHAR_M_IV, and QLIKE-trained neural models perform poorly in the saved run. This means the large-universe extension does not currently show a stronger GNN improvement just because the graph has more nodes. It instead suggests that large-universe GNN training is more sensitive to optimization, scaling, and possibly over-smoothing.

8 Current Scope Gaps

The current local outputs support the one-day Zhang-style analysis. The following pieces cannot be reconstructed exactly from the saved forecasts alone:

- one-week and one-month horizon tables, because those require horizon-specific targets and reruns;
- Zhang’s exact SPY regime split, unless a SPY RV series is added and aligned;
- exact hidden-state MAD from Zhang’s Figure 7, because hidden representations were not saved;
- the smaller-validation robustness table, because that changes the rolling train-validation split.

The generated CSV layer records these limitations explicitly in `outputs/paper_ready_20260617/zhang_style_sta`

9 Generated Files

The complete machine-readable results are stored under `outputs/paper_ready_20260617/zhang_style_statistics` and under each universe’s `zhang_style_statistics/` folder. The most important cross-universe files are:

- `cross_universe_loss_ratio_summary.csv`
- `cross_universe_mcs_rollup.csv`
- `cross_universe_pairwise_qlike_dm_summary.csv`
- `cross_universe_fvu_by_regime_vs_HAR_M.csv`
- `cross_universe_ticker_alignment_audit.csv`
- `cross_universe_graph_rollup.csv`

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